

Hello everyone. This is Sania Sadaf and welcome back to my series. This is Part 3 of BLA 3 in my AWS Certified Machine Learning series. Part 1 explained model selection, and Part 2 covered training and customization. This final presentation focuses on determining whether a model is good enough to use and how its behavior can be improved over time.

I will discuss baselines, traditional machine learning metrics, generative AI evaluation, explainability, reproducible experiments, human review, shadow testing, drift detection, and RAG monitoring.

Evaluation connects technical model behavior to business risk. A model can have a strong average score and still fail on important groups, rare cases, unsafe prompts, or time-sensitive situations. For that reason, evaluation should use several kinds of evidence. The team needs suitable metrics, representative test data, human judgment when meaning matters, and continued monitoring after deployment.

A useful evaluation begins with success criteria. The team should define what the system must do, which mistakes matter most, and what level of performance is acceptable.

Technical quality depends on the task. Classification, regression, text generation, and retrieval require different measurements.

Operating behavior also matters. The model must respond within the required time, handle expected traffic, remain available, and stay within budget.

The test set should represent real use, including important subgroups and difficult edge cases.

Classification metrics describe different kinds of correctness. Accuracy measures the percentage of predictions that are correct overall, but it can be misleading when one class appears much more often than another. Precision measures how often positive predictions are correct. Recall measures how many actual positive cases the model finds. F1 score combines precision and recall into one value.

A confusion matrix separates true positives, true negatives, false positives, and false negatives. This helps the team see which error is occurring.

Regression metrics measure numerical errors. Mean absolute error gives the average size of the error. Root mean squared error gives more influence to large errors. R-squared describes how much variation the model explains, but it does not show whether an error is acceptable to the business.

Generative AI evaluation is difficult because several different answers may be acceptable. Automated metrics can still help.

These metrics have limits. A response can use different words and still be correct, or it can repeat reference words while giving a misleading answer. Human evaluation is therefore important for correctness, relevance, clarity, usefulness, tone, and safety.

An LLM-as-a-judge approach uses another model to score outputs.

Generative AI evaluation should include normal requests, edge cases, adversarial prompts, sensitive topics, and different user groups.

Explainability helps users and developers understand why a model produced an output. Global explanations describe the features and patterns that influence the model overall. They can show which variables usually matter and whether the model depends on unexpected information.

Local explanations focus on one prediction. For example, a local explanation may show which features increased or decreased a risk score for a particular record.

The explanation method should match the model and audience. A data scientist may need detailed feature contributions, while a business user may need a plain-language reason and the limits of that reason.

Explanations should not be treated as perfect proof. Some methods approximate complex model behavior and can be unstable. The team should test explanation quality, protect sensitive information, and check whether the explanation could create false confidence.

A reproducible experiment changes a clearly defined factor and records the result. The team begins with a baseline, then changes a model, dataset, prompt, retrieval setting, or hyperparameter.

Experiment tracking records the data version, code, model, prompts, configuration, metrics, and artifacts. MLflow on SageMaker AI can support experiment tracking for machine learning workflows.

Human evaluation should use clear scoring instructions and representative reviewers. Reviewers may rate correctness, relevance, style, safety, or preference.

Experiments should report both improvements and tradeoffs. A candidate may improve quality while increasing latency or cost.

Offline evaluation cannot reproduce every production condition. Shadow testing sends a copy of real traffic to a candidate model without allowing that model to affect the user.

Drift monitoring detects changes over time. Data drift means the input distribution has changed.

RAG systems need additional monitoring. Retrieval evaluation should measure whether the system finds relevant documents and whether those documents support the answer.

To conclude Part 3 and BLA 3, evaluation should continue throughout the model lifecycle. The team begins with a baseline and clear success criteria.

Traditional machine learning uses classification or regression metrics, while generative AI requires automated measures, human evaluation, safety testing, and checks for grounding.

My main takeaway from Domain 2 is that model development includes choosing, training, evaluating, and improving a solution. A model is ready only when its quality, cost, latency, safety, and operating behavior meet the real requirement. The next BLA will move into deployment and orchestration of machine learning and AI workflows.